

Financial Vulnerability in Canadian Households

A statistical and machine-learning analysis of Statistics Canada's 2023 Survey of Financial Security

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github.com/mbhatti48/Data-Project/financial-vulnerability-canada

1. OBJECTIVE

About 4% of Canadian families report skipping a payment because of financial hardship. Against the backdrop of Canada's 2025-26 technical recession, this project asks two questions about those families: **what separates them from everyone else, and can they be predicted?**

"Vulnerable" is defined as a family that skipped a payment due to financial difficulty. This is the survey's own recorded distress signal, which avoids inventing an arbitrary threshold and matches how banks and the Bank of Canada measure household trouble. It also keeps every balance-sheet column available as a predictor.

2. DATA AND PREPARATION

The analysis uses Statistics Canada's 2023 Survey of Financial Security Public Use Microdata File: **16,241 economic families**, from which 23 of the original 102 variables were selected (income, assets, debts, payment-stress flags, and demographics).

The survey stores most answers as numeric codes rather than values, so cleaning was handled code by code: valid skips (where a question did not apply) were folded into meaningful "No" responses; genuinely missing answers were imputed using the survey weights, via a weighted median for education and a weighted mode for retirement status; and a disguised sentinel value of 99,999,999 in market income was identified and replaced with a weighted median. Where a missing answer carried its own meaning, on a sensitive question, it was kept as an explicit "unknown" category rather than guessed at.

Scope note: the survey covers housed families in the ten provinces and excludes people who are homeless, institutionalized, or living on reserves, so it understates the country's most extreme vulnerability.

3. EXPLORATORY ANALYSIS AND HYPOTHESIS TESTING

Exploration showed every dollar column heavily right-skewed, a near-duplicate pair (total assets and net worth correlated at 0.98), and a clear signal: families showing payment stress sat visibly lower on net worth. Two hypotheses followed, each tested at a 0.05 significance level.

- **The wealth gap (Welch's t-test).** Vulnerable families hold significantly less net worth than everyone else: $t = -17.21$, $p = 8.31e-57$. Welch's version was used because the two groups have very different spreads. The gap is real, not sampling noise.
- **The education pattern (chi-square test).** Hardship rates differ across education levels: chi-square = 27.48, $p = 4.67e-06$. The pattern was not the expected downward staircase but an **arch**, with hardship peaking at the middle levels (3.4%, 4.9%, 5.0%, 3.3%). A two-column test cannot say why, which set up a question for the model.

4. MODEL DESIGN

One feature was engineered: **debt-to-income**, total debt divided by after-tax income, the household-vulnerability ratio the Bank of Canada tracks. Its edge cases were handled explicitly rather than dropped, with no-debt families assigned a true value of zero and near-zero-income artifacts capped at the 99th percentile.

The two payment-stress flags closest to the target were excluded to prevent leakage, and total assets was dropped for its 0.98 correlation with net worth, which would have made the interpretation unstable. That left **18 features**. Category columns were one-hot encoded and numeric columns scaled, and the data was split into training and test sets **before** those settings were learned, so nothing from the test set could leak into preparation.

Two models were trained and compared: a **logistic regression** (interpretable, the industry standard for credit-risk scorecards) and a **random forest** (the stronger predictor on tabular data). Both were given balanced class weights so the rare 4% could not be ignored, and both were tuned by cross-validated grid search on the training data only.

5. MODEL EVALUATION

Both models were scored once, on a held-out test set. **Accuracy was rejected as a grade:** with only 4% vulnerable, a model that calls everyone safe scores 96% and is useless.

Metric	Logistic Regression	Random Forest	Why it matters here
Accuracy	0.69	0.85	Not meaningful on a rare class
Precision	0.09	0.15	Low: most flagged families are false alarms
Recall	0.72	0.54	The priority: missing a vulnerable family is the costly error
F1	0.16	0.23	Used to tune each model
AUC	0.79	0.83	The fair way to compare the two models' ranking

Test-set performance at the default 0.5 decision threshold.

The random forest ranked families better (AUC 0.83), so it was kept. Because it is cautious about a rare class, its **decision threshold was lowered from the default 0.5**, trading precision for recall deliberately.

Decision threshold	Recall	Precision	Share of families flagged
0.5 (default)	0.54	0.15	~15%
0.3 (chosen)	0.80	0.11	~30%
0.2	0.91	0.09	~42%

At 0.3 the model **catches 4 out of 5 vulnerable families** while flagging a manageable share. Dropping to 0.2 buys little and flags nearly half the population. The result is a **wide-net screening tool, not a precision instrument**: it identifies a genuinely higher-risk group but cannot pinpoint exactly who is vulnerable. That is the appropriate trade when missing a vulnerable family costs more than a follow-up on a false alarm.

6. WHAT DRIVES VULNERABILITY

The logistic model's odds ratios and the forest's feature importances were read side by side. Two very differently built models independently ranked the same features at the top, which is strong evidence the signal is real.

Feature	Odds ratio (logistic)	Forest importance	Effect on vulnerability
Credit-card debt	1,000+	0.19	Raises risk, steepest in the model
Net worth	≈0	0.16	Protects, the strongest safeguard
Debt-to-income	1.72	0.12	Raises risk
Total debt	-	0.10	Raises risk
Student-loan debt	61	-	Raises risk
Age of main earner	0.14	-	Protects
Education	0.72	outside top 10	Negligible once age and wealth are held constant

Odds ratios above 1 raise the odds, below 1 lower them. Numeric features were scaled to a 0-1 range, so each multiplier describes moving from the bottom to the top of that feature's range, not a per-dollar effect. Importances carry no direction.

- **Every kind of debt raises risk, credit-card debt most steeply** (also the forest's single most-used feature). Student loans and other consumer debt follow.
- **Net worth, income, and age protect strongly.** Older, wealthier families with income coming in are far safer, everything else held equal.
- **The education arch was mostly age in disguise.** With age and net worth in the model at once, education's effect nearly vanished (odds ratio 0.72), and it did not crack the forest's top ten. The seemingly safe least-educated group was protected by being older and wealthier, not by low education. This is precisely what the chi-square could not untangle on its own.

The profile: the vulnerable family carries consumer debt, credit-card debt above all, with little net worth behind it, modest income, and a younger main earner. Demographics barely move the needle once the balance sheet is known. Vulnerability is not about who a family is; it is about what they owe versus what they have to fall back on.

7. BENCHMARK CHECK: THE BANK OF CANADA'S DEBT LINE

As an extension, the findings were tested against a real institutional benchmark. The Bank of Canada has long flagged a household as highly indebted when its debt exceeds **350% of income**. Does that official line catch the families who actually miss payments?

Only partly. Families above the line do miss payments at nearly twice the rate of everyone else (6.8% vs 3.8%), so it detects real risk. But of the 672 families who actually broke, **it missed 545, about 81%**. The reason reconciles with the model: the 350% line runs on **total** debt, roughly 79% mortgage for the families it flags, so it mostly identifies asset-rich homeowners with median net worth above \$1M, exactly the families the model reads as safe. The families who actually break carry modest total debt but thin net worth (median roughly \$221K against \$905K overall) and the consumer debt the model singled out. Debt does drive vulnerability; this line simply measures the wrong slice of it.

Weighted to the national population, 13.4% of families fall above the line, close to the Bank's own published share and consistent with this project's use of after-tax rather than gross income, which lifts every family's ratio slightly.

8. LIMITATIONS

- The survey cannot see the most vulnerable Canadians, who sit outside its frame entirely.
- The target captures families already missing payments, not families quietly draining savings to stay afloat.
- The exploratory charts are unweighted and describe the sample rather than the population; the imputations do use the survey weights.
- Imputation fills and the debt-to-income cap were computed on the full dataset before the split. The effect is negligible, but a stricter build would compute both from the training half only.

9. CONCLUSION

Both questions are answered. **What separates vulnerable families is the balance sheet**, consumer debt set against thin reserves, not demographics. **They can be predicted, as a screening tool:** a tuned random forest with its threshold lowered to suit a rare class catches 4 of 5 vulnerable families (AUC 0.83), casting a wide net by design. Held against the Bank of Canada's own debt benchmark, the model's reading holds up and exposes a real limitation in the benchmark: a single debt ratio is a useful screen, but finding vulnerable families takes the whole balance sheet.

10. TOOLS AND SOURCES

Built in Python: pandas and NumPy for the data work, Matplotlib and Seaborn for visualization, SciPy and statsmodels for the statistical tests, and scikit-learn for preprocessing, modelling, and evaluation. The annotated notebook and data dictionary are in the repository.

Data: Statistics Canada, *Survey of Financial Security 2023* Public Use Microdata File, Catalogue no. 13M0006X, under the Statistics Canada Open Licence. **Benchmark:** Bank of Canada, *Financial System Review 2019*, source of the 350 percent threshold; the Bank has since shifted its headline measure, but that line matches the ratio computed here.